# ZHEXIN(TONY) WU

Rämistrasse 101, Zürich, Schweiz | zhexwu at student dot ethz dot ch | https://10258392511.github.io/zhexinwu.github.io/

## EDUCATION

**Department of Mathematics, ETH Zürich** — **Doctoral Student, Sep 2026 – Present**
**Research: Optimal Execution, Market Microstructure & Reinforcement Learning**
**Supervisor: Prof. Patrick Cheridito**

**Management, Technology and Economics, ETH Zürich** — **Master of Science with Distinction, Sep 2023 – Sep 2026**
**Focus: Quantitative Finance** — **GPA: 5.85/6.00**

**Information Technology and Electrical Engineering, ETH Zürich** — **Master of Science with Distinction, Sep 2021 – Jul 2023**
**Major: Signal Processing and Machine Learning** — **GPA: 5.76/6.00**

**Electrical and Computer Engineering, University of Michigan – Ann Arbor** — **Master of Science Program, Sep 2020 – May 2021**
**Major: Signal and Image Processing and Machine Learning (SIPML)** — **GPA: 4.00/4.00**

**Shen Yuan Honors College, Beihang University** — **Bachelor of Engineering, Sep 2015 – Jul 2019**
**Major: Electronic and Information Engineering** — **GPA: 3.80/4.00, Rank: 15/209**

## CERTIFICATIONS

**CFA Program** — **Passed Level III, Oct 2026**

## INTERNSHIPS

**IPO Internship** — **Analyst**
*Sinolink Securities[code]* — *Jun 2024 to Sep 2024*
- Developed scraper programs to collect and analyze the latest data on IPO statuses, prospectuses and inquiry letters from public websites
- Developed a custom GPT to construct structured industry and company analysis reports

**Quantitative Internship in Factor Investment** — **Quantitative Analyst**
*China Securities[code]* — *Feb 2024 to Mar 2024*
- Constructed CSI 300 and CSI 500 factor database using Tushare data terminal
- Developed trading strategy with information-coefficient-weighted factor using Zipline framework

## RESEARCH EXPERIENCE

**Multi-Agent Reinforcement Learning for Market Microstructure and Optimal Execution**
*With Dr. Jean-Loup Dupret and Prof. Patrick Cheridito[paper]* — *Feb 2025 to Present*
- Developed a novel multi-agent reinforcement learning (MARL) framework based on the Kyle model to study strategic interactions among heterogeneous groups of market participants and market efficiency through limit order book (LOB) dynamics
- Extended the Almgren-Chriss model with RL-based agents and an endogenous price process for optimal execution
- Refactored ABIDES-Gym to enable multi-agent training and asynchronous execution in a simulated LOB market

**In-Vivo Fetal Thalamus Parcellation** — **Project Collaborator & Software Developer**
*With Dr. Hui Ji and Prof. András Jakab[paper][code]* — *Mar 2023 to Dec 2023*
- Unsupervised fetal thalamus functional group recognition with clustering algorithms
- Feature engineering using coordinate and spherical harmonic coefficient from diffusion MRI
- Optimized for high reproducibility between scan and re-scan data with image registration and Bayesian optimization
- Implemented scree plot, silhouette score and gap statistic for choosing optimal number of clusters

**MR Image Reconstruction with Diffusion Models and Implicit Neural Representations** — **Master's Thesis**
*Advised by Dr. Valery Vishnevskiy and Prof. Sebastian Kozerke*
*[paper][slides]code (diffusion models)[code (implicit NR)]* — *Oct 2022 to Jun 2023*
- Developed score-based diffusion models for 2D and 2D + time CMR image reconstruction with high acceleration rates
- Proposed proximal mapping for complex-valued image reconstruction algorithm with score-based diffusion models
- Conducted extensive ablation study on type of algorithm (sampling or MAP), choice of prior (model-based or data-driven) and patching-artefacts removal strategies
- Developed implicit neural representations for 2D, 2D + time and higher dimensional CMR image reconstruction

**Sequence Models in 3D MRI Brain Metastases Detection** — **Semester Thesis**
*Advised by Prof. Ender Konukoglu [paper][slides][code]* — *Apr 2022 to Jul 2022*
- Task: Brain metastases detection in $T_1$-weighted 3D MR images
- Revealed statistical significance of sequence models by training simple continuous patch-sequence classifier
- Reduced false positives by incorporating active learning
- Investigated the use of deep reinforcement learning agent

**Effects of Data Augmentation and Semi-Supervised Learning in Domain Generalization** **Semester Thesis**
*Advised by Prof. Ender Konukoglu [paper][slides][code]* *Oct 2021 to Mar 2022*

- Task: 2D MRI cardiac image segmentation under domain-shift
- Investigated using consistency loss as both auxiliary loss for training and proxy loss for test-time adaptation
- Proposed a new inversion-based data augmentation
- Conducted rich experiments on different data augmentation schemes
- Investigated the use of meta-learning

**Manifold Optimization & Neural Collapse** **Research Assistant**
*Advised by Prof. Qing Qu[paper] [code]* *Jan 2021 to Jun 2021*

- Created a smooth introduction to Riemannian submanifold optimization, with geometric illustration and concrete algorithmic examples.
- Participated in project *A Geometric Analysis of Neural Collapse with Unconstrained Features* and proof-read the script.
- Carried out experiments on neural collapse of CNN's last layer with oblique-manifold constraints on features and/or layer weights.

## PROJECTS

**Walmart Sales Prediction** **Course Project**
*[executive summary][slides][code]* *Mar 2024 to May 2024*

- Developed machine learning pipeline following CRISP-DM, with an emphasis on explanatory model analysis
- Created executive recommendation from model prediction and variable importance

**Model Fusion for Medical Image Segmentation via Optimal Transport** **Course Project**
*[paper][code]* *Sep 2022 to Jan 2023*

- Extension of model fusion via optimal transport from cascade-like architecture (VGG, ResNet &etc) to more general architecture whose computational graph has richer adjacency structure
- Proposed model fusion approaches for prevalent U-Net and Transformer architectures used for medical image segmentation
- Comprehensive ablation study on fusion weighting scheme for typical medical data scenarios including distributed learning and domain-shift issue

**Permutation-Invariant Variational Network for 2D + Time Cardiac Image Reconstruction** **Course Project**
*[paper][code]* *May 2022 to Jun 2022*

- Implemented permutation-invariant variational network in PyTorch
- Implemented 2D and 3D variational network to make comparison with permutation-invariant variational network
- Re-organized derivation of optimization steps for loop-unrolling

**Monitoring Social Distancing and Mask Wearing** **EECS 504 Course Project**
*[paper] [video][code]* *Nov 2020 to Dec 2020*

- Selected the topic and created the framework of the pipeline
- Responsible for human instance segmentation and face detection; proposed a chin locator based on the results of instance segmentation via Mask R-CNN
- Augmented face dataset with Gaussian Pyramid and integrated multiple datasets
- Improved the performance of mask classifier by designing a upsampler based on DCGAN with a group member
- Created prototype of social distance circle mapping using homography with a group member

**Michigan Tourist Guide** **SI 507 Course Project**
*[report] [video] [code]* *Mar 2021 to Apr 2021*

- Gathered data from multiple sources with web APIs and web crawling with BeautifulSoup4.
- Created a full-stack web app with Flask with rich interactive functionalities.

**The 13th Mathematical Modeling Contest of Beijing Normal University** **Team Leader, Second Prize**
*Advised By Assoc. Prof. Yingzhe Wang* *Apr 2017 to May 2017*
**Project: Tourist-flow Prediction at South Luogu Lane in Beijing, China**

- Constructed cellular automata model to simulate the changes in tourist-flow with MATLAB
- Later improved theoretical basis of the paper for the contest by constructing a model based on PDEs (Partial Differential Equations) with varied boundary conditions (compared to the original map); the polished work as the final course project achieved 1/209 of the whole class

## SELECTED ONLINE COURSES AND PROJECTS

A full list of online courses & projects can be viewed here

- ***Sparse Representations in Signal and Image Processing (by Technion)*** *Jul 2021 to Aug 2021*
  - o ***Topics:*** Theoretical analysis of $P_0$ problem and guarantee of pursuit algorithms' stability; greedy pursuit algorithms and $L_1$ relaxation; FISTA and K-SVD algorithms(Dictionary Learning); application of sparseland model in image denoising, morphological components analysis(MCA), super resolution & etc.
  - o ***Projects in Python:*** OMP & Basis Pursuit for solving $P_0$; Image Reconstruction with OMP; Unitary Dictionary Learning; Image Denoising with Dictionary Learning; Image Deblurring

## AWARDS

- National Scholarship for 2016-2017 academic year
- Merit Student for 2015-2016, 2016-2017 and 2017-2018 academic year
- First Prize of Learning Merit Scholarship for 2015-2016 and 2016-2017 academic year
- University Excellent Student for 2016-2017 academic year
- Second prize of The 13th Mathematical Modeling Contest of Beijing Normal University
- Third Prize of Academic Competition Scholarship
- First Prize of Excellent Social Work
- The 27th and 28th National High School Student Chemistry Competition, Second Prizes

## SKILLS SUMMARY

**Programming Languages** Python, C/C++, Julia, MATLAB, Bash Script, Java, LaTeX
**Quant Frameworks** Backtrader, VNPY, Zipline, Qlib
**Machine Learning** PyTorch, TensorFlow, JAX, RLlib, OpenCV
**Statistical Analysis** R, Stata, SPSS, Excel VBA
**Financial Data Terminals** LSEG Workspace, Wind Financial Terminal
**Database Management** SQL, PySpark
**Web Development** HTML5, CSS3, JavaScript, BootStrap4, Django, Flask, Ruby on Rails
**Hobbies** Classical Piano, English Literature
**Languages** Chinese (Native), English (Fluent), Deutsch (Anfänger)

## GRADUATE COURSES AT ETH ZÜRICH

401-6624-11L Applied Time Series
363-0392-00L Strategic Management
363-1077-00L Entrepreneurship
401-3929-00L Financial Risk Management in Social and Pension Insurance
363-0421-00L Management of Digital Transformation
363-0389-00L Technology and Innovation Management
401-3629-00L Quantitative Risk Management
363-0584-00L International Monetary Economics
363-1080-00L Responsible Leadership
401-3915-73L Machine Learning in Finance and Insurance[code]
401-4623-00L Time Series Analysis
363-0585-00L Intermediate Econometrics
363-1021-00L Monetary Policy
363-0560-00L Financial Management
363-0570-00L Principles of Econometrics
363-1098-00L Business Analytics[code]
363-0515-00L Decisions, Markets, and Games
363-0575-00L Economic Growth, Cycles and Policy
363-1063-00L Academic Writing Course
363-0711-00L Accounting for Managers
363-0503-00L Principles of Microeconomics
363-0565-00L Principles of Macroeconomics
363-0541-00L Economic Dynamics and Complexity
363-1004-00L Operations Research
363-0453-00L Strategic Supply Chain Management
363-0341-00L Introduction to Management
363-0403-00L Introduction to Marketing

263-3210-00L Deep Learning[code]
263-5210-00L Probabilistic Artificial Intelligence[code]
401-4944-20L Mathematics of Data Science
851-0252-15L Network Analysis
252-0220-00L Introduction to Machine Learning [code]
227-0391-00L Medical Image Analysis
227-0424-00L Model- and Learning-Based Inverse Problems in Imaging[code]
227-0948-00L Magnetic Resonance Imaging in Medicine
227-0449-00L Seminar in Biomedical Image Computing
401-3901-00L Linear & Combinatorial Optimization
401-3621-00L Fundamentals of Mathematical Statistics
227-0447-00L Image Analysis and Computer Vision
MOB-001 Basic German 1; A1: Intensive Course / Sprachenzentrum der UZH und der ETH Zürich

## GRADUATE COURSES AT UMICH

EECS 501 Probability and Random Processes
EECS 559 Optimization Methods in Signal Processing and Machine Learning
SI 507 Intermediate Programming[code]
EECS 551 Matrix Methods for Signal Processing, Data Analysis and Machine Learning
EECS 504 Foundations of Computer Vision[code]

## AUDITED COURSES

401-3904-22L Convex Optimization
EECS 598-005 Deep Learning for Computer Vision[code]